Data Lake Value Proposition

Medical Data Processing Company

EDWAED AMANKWAH

CONFIDENTIAL

DATA LAKE

VALUE PROPOSITION

Medical Data Processing Company

Enterprise Data Architecture Proposal | 2024

AGENDA

Today's Discussion

01 **What is a Data Lake**

02 Components of a Data Lake

03 Data Lake vs Data Warehouse

04 Business Value of Data Lake

05 Proposed Architecture

What is a Data Lake

Executive Summary

A Data Lake is a centralized, scalable repository that stores structured, semi-structured, and unstructured data in its native format at any scale. Unlike traditional databases, a Data Lake uses a "schema-on-read" approach — meaning data is stored as-is and structure is applied only when it is read for analysis. This enables organizations to collect all data first and determine its value later, dramatically accelerating innovation and insights.

STORE EVERYTHING

All data types — structured, semi-structured, unstructured — in one place. No data left behind.

SCALE INFINITELY

Horizontally scalable across commodity hardware. Grows with your business at low incremental cost.

ANALYZE FREELY

Multiple analytics engines (SQL, ML, streaming) against the same data — no data movement required.

Components of a Data Lake

Think About: What elements are needed to make a data lake? Each layer serves a distinct purpose in the data journey.

01

Ingestion Layer

Apache NiFi · Kafka · Sqoop · Debezium

Collects data from all sources (FTP, APIs, streams, databases) in real-time. Handles 15M+ files/day from 8,000 facilities.

02

Storage Layer

HDFS · Parquet · Apache Hudi · Apache Atlas

Centralized, scalable repository with Raw, Staging, Curated & Archive zones. Unlimited retention, fault-tolerant with 3x replication.

03

Processing Layer

Apache Spark · Kafka Streams · Airflow · Hive

Transforms raw data into insights via batch, stream, and interactive query. Metadata-driven design replaces 70+ custom scripts.

04

Serving Layer

Superset · Impala · Presto · MLflow · REST APIs

Delivers curated data to dashboards, reports, APIs, and ML models. Enables multiple use cases from one data source without duplication.

05

Security & Governance

Apache Ranger · Atlas · Kerberos · Knox

HIPAA-compliant RBAC, encryption at rest (AES-256), in-transit TLS, data lineage, and audit trails across all layers.

Data Lake vs Data Warehouse

Why a Data Lake is the right choice for Medical Data Processing Company

DATA WAREHOUSE	DATA LAKE ✓
Schema-on-Write (rigid)	✓ **Schema-on-Read (flexible)**
Structured data only	✓ **Structured + Semi + Unstructured**
Vertical scaling (expensive)	✓ **Horizontal scaling (commodity HW)**
Fixed, high storage costs	✓ **Low-cost, tiered storage**
Proprietary SQL databases	✓ **100% Open-source ecosystem**
Single use case per dataset	✓ **Multiple engines on same data**
Limited ML integration	✓ **Native ML/AI/TensorFlow support**
Days to onboard new source	✓ **Hours with metadata-driven design**

Business Value of Data Lake

8,000+
Facilities Served Without Disruption

15M
Files/Day Processed

99.9%
Target Uptime (vs. crashes today)

40-60%
Storage Cost Reduction (tiering)

Eliminate System Downtime

Replace the single-point-of-failure SQL Server with a distributed HDFS cluster (3x replication). Node failures become routine, not disasters. Zero planned downtime for maintenance.

Real-Time Patient Insights

Move from nightly-batch-only to near-real-time data processing. Bed availability, patient metrics, and compliance dashboards update in minutes — not the next morning.

Unlimited, Affordable Scalability

Horizontal scale-out on commodity hardware grows storage and compute independently. 15-20% annual data growth is sustainable with no performance degradation.

Eliminate Data Silos

One centralized data lake replaces fragmented exports across departments. All 8,000 facility datasets are discoverable, accessible, and governed from a single platform.

Accelerate Innovation with ML/AI

Machine learning models (TensorFlow, Spark MLlib) run directly on the data lake. Predict readmissions, detect anomalies, and drive clinical intelligence — without moving data.

Metadata-Driven Automation

Replace 70+ hand-crafted SSIS scripts with a single parameterized framework. Onboard new data sources in hours, not weeks. Radically reduce operational maintenance overhead.

WHY NOW?

The Case for Immediate Action

CRISIS

System Crash

Last week's ETL surge crashed SQL Server — 8,000 facilities offline for hours. This WILL happen again, more severely, as data grows.

GROWTH

Data Explosion

8TB today growing at 15-20% YoY. SQL Server capacity (12TB, 70% full) will be exhausted within 12-18 months. Clock is ticking.

RISK

HIPAA Exposure

Single-node SQL Server with manual backups creates unacceptable HIPAA compliance risk. A data breach or prolonged outage carries severe regulatory penalties.

OPPORTUNITY

Competitive Edge

Competitors are building ML-powered predictive analytics. Medical Data Processing must evolve from batch-reporting to real-time intelligence or risk losing market position.

Proposed Data Lake Architecture(Open Source Stack)- Simplified Version

Data Lake Solution Architecture Diagram

On-Premise Hadoop Ecosystem · HIPAA Compliant · Horizontally Scalable

CLIENTS

- **Medical Facility** — 8,000 sites
- **FTP / SFTP** — File Transfers
- **REST APIs** — External Systems
- **Legacy SQL** — SQL Server Migration
- **File Formats** — CSV · XML · TXT; 15M files/day
- **Data Volume** — 77,000 zip files/day · 3,500/hr; 15–20% YoY growth
- **End Users** — Analysts · Engineers · Customers; BI · Ad-hoc · Reporting

INGESTION — ① INGESTION LAYER

- **Apache NiFi** — Visual Dataflow · FTP Polling; Gzip Decompression · Validation
- **Apache Kafka** — Message Queue · 3,500 files/hr; Topic Partitioning · Replay
- **Debezium** — Change Data Capture (CDC); SQL Server Logs · UPSERT
- **Apache Sqoop** — Bulk DB Import · SQL → HDFS; Parallel Import · Incremental
- **Apache Flume** — Log & File Collection; Agent-Based · Reliable Delivery
- **Apache Airflow** — Orchestration · Scheduling; Pipeline Monitoring

STORAGE — ② STORAGE LAYER — HDFS (Replication Factor: 3)

- **Raw Zone** — Landing Area · Immutable; Parquet / ORC · Audit Trail
- **Staging Zone** — Cleaned & Validated; Apache Hudi · UPSERT · CDC
- **Curated Zone** — Gold · Analytics-Ready; Parquet · Partitioned · 5yr+
- **Archive Zone** — Cold Storage · HIPAA 7yr+; SSD → HDD → JBOD Tiering
- **Metadata Store** — Apache Atlas · Data Catalog; Lineage · PII/PHI · Hive Metastore

PROCESSING — ③ PROCESSING LAYER

- **Batch Processing** — Apache Spark (PySpark) · YARN; Metadata-Driven ETL · Replaces SSIS
- **Stream Processing** — Kafka Streams · Apache Flink; Near-Real-Time · Sub-minute Latency
- **Ad-hoc & CDC** — Hive · Impala · Presto / Trino; Federated SQL · Time-Travel Queries
- **ML & Analytics** — Spark MLlib · TensorFlow · MLflow; Risk Scoring · Experiment Tracking

SERVING — ④ SERVING LAYER

- **SQL Reports** — Hive · Impala; Daily · Weekly · Compliance
- **Dashboards** — Apache Superset · Grafana; PowerBI · Tableau · Real-time
- **Web Portal & APIs** — FastAPI · Customer Self-Service; Bed Availability · Load Balanced
- **ML Model Serving** — MLflow REST · Predictions; Risk Scores · New Capability
- **Ad-hoc Analytics** — Presto / Trino · Jupyter Notebooks; Federated SQL · Eliminates Silos

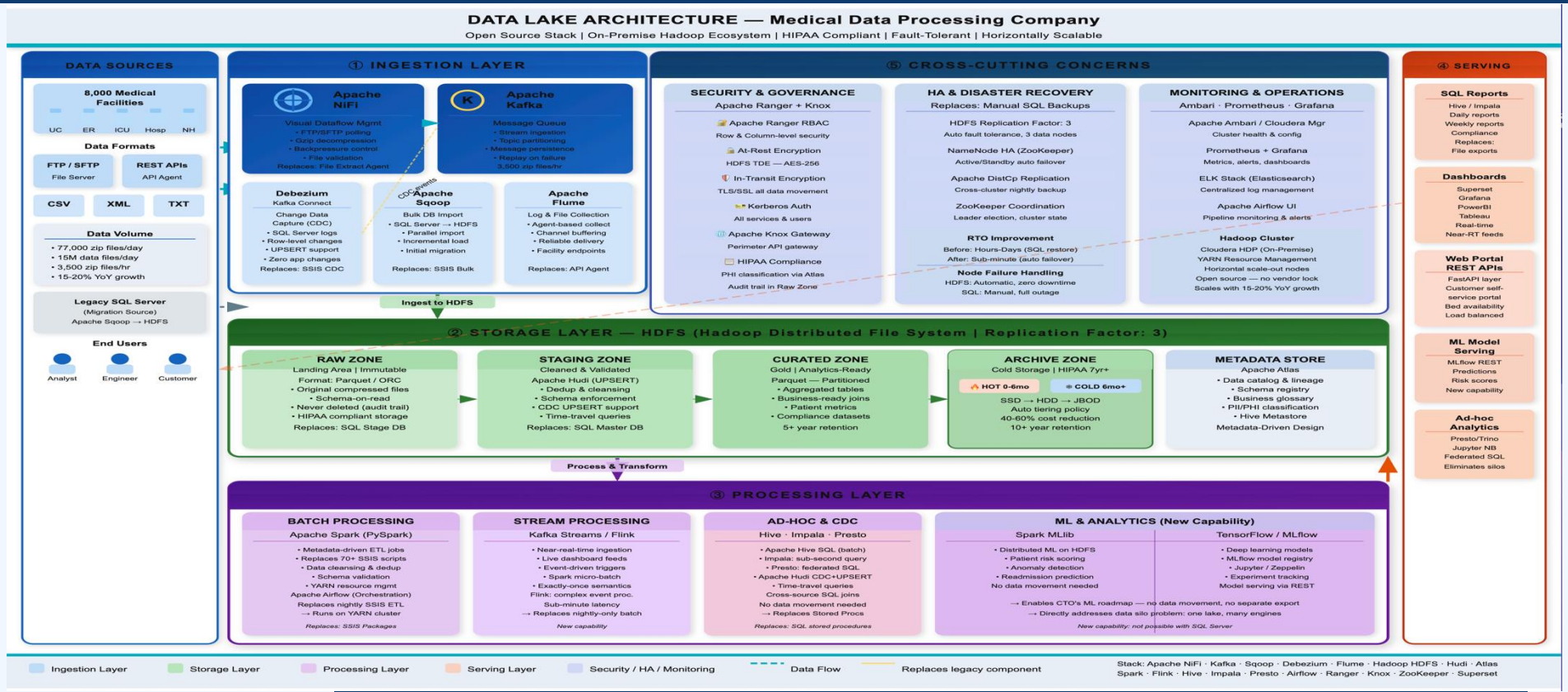
CROSS-CUTTING — ⑤ CROSS-CUTTING CONCERNS — Spans All Layers

- **Security & Governance** — Apache Ranger (RBAC) · Knox Gateway; Kerberos Auth · AES-256 · TLS/SSL; HIPAA Compliance · PHI Classification
- **HA & Disaster Recovery** — HDFS Replication Factor: 3; NameNode HA · ZooKeeper Failover; Apache DistCp · Sub-minute RTO
- **Monitoring & Operations** — Ambari / Cloudera Manager; Prometheus · Grafana · ELK Stack; Airflow UI · Pipeline Alerts
- **Hadoop Cluster** — Cloudera HDP · On-Premise; YARN Resource Management; Horizontal Scale-out · No Vendor Lock

Ingestion · Storage · Processing · Serving · Cross-Cutting · ---- Data Flow

Stack: NiFi · Kafka · Debezium · Sqoop · Flume · HDFS · Hudi · Atlas · Spark · Flink · Hive · Impala · Presto · Airflow · Ranger · Knox · ZooKeeper · Superset

Proposed Data Lake Architecture — Open Source Stack



CROSS-CUTTING: Apache Ranger (Security + RBAC) • Apache Atlas (Governance) • ZooKeeper (HA) • Prometheus/Grafana (Monitoring) • Kerberos (Auth)

Industry Success Stories — Data Lake Proven at Scale

Other leading organizations have already embraced Data Lake strategies — with measurable results:

Netflix

Media & Tech

Built a multi-petabyte Data Lake using Apache Spark + Iceberg. Processes billions of events daily for 230M+ subscribers. Eliminated data silos across 100+ microservices.

Uber

Ride-sharing

Invented Apache Hudi to solve CDC/UPSERT at scale on Hadoop. Processes 1TB+ per hour. Enabled real-time driver-rider matching analytics that power their core business.

LinkedIn

Professional Network

Invented Apache Kafka. Built a unified Data Lake (DataHub) serving 100M+ members. Processes trillions of events per day for relevance, ML, and compliance.

NYU Langone Health

Healthcare

HIPAA-compliant medical Data Lake for clinical analytics. Enabled real-time ICU monitoring dashboards. Demonstrably reduced patient mortality rates through data-driven care decisions.

Mayo Clinic

Healthcare Research

Integrated EMR Data Lake enabling genomic research and AI-assisted diagnostics. Processes billions of clinical data points. Currently a global leader in data-driven medical research.

UDACITY

THANK YOU

Let's build the future of medical data analytics together.

Questions & Discussion